

Kwasi Asante-Gyimah

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EDUCATION

Fordham University

August 2024 – May 2028

B.Sc. in Mathematics & Computer Science

New York, NY

- **Relevant Coursework:** Computer Algorithms, Data Structures, Discrete Mathematics
- **Social Chair, Society for Industrial and Applied Mathematics (SIAM):** Organized chapter events, collaborated with SIAM chapters across other universities, and managed the chapter website

EXPERIENCE

Fordham University – Department of Computer Science

May 2026 – Present

Undergraduate Research Assistant

New York, NY

- Conducting research under Prof. Xiaolan Zhang applying **reinforcement learning** to scheduling problems in quantum networks
- Implementing a **value iteration** solver for a Markov Decision Process (MDP) formulation in **Python**, producing an exact optimal policy benchmark to evaluate and compare existing RL algorithm performance

PROJECTS

RAG Pipeline | *Python, FastAPI, OpenAI, Anthropic, ChromaDB, BM25, LangChain, Docker*

- Built a production-ready RAG pipeline with hybrid dense + sparse search fused via **Reciprocal Rank Fusion**, delivering semantically accurate, citation-grounded answers across OpenAI and Anthropic models
- Engineered a **FastAPI** backend integrating OpenAI embeddings, **ChromaDB** vector storage, and multi-provider LLM dispatch; implemented a four-stage document pipeline (ingest → index → retrieve → generate) with idempotent chunk upserts for safe re-ingestion of updated documents
- Designed a parallel retrieval architecture firing dense and sparse (**BM25**) searches simultaneously with source-level citation enforcement on every generated claim for full answer auditability

Accentr: AI Pronunciation Trainer | *TypeScript, Next.js, Anthropic Claude API, ElevenLabs, PostgreSQL, Prisma*

- Built a full-stack language learning app where users record speech and receive **phoneme-level pronunciation feedback** driven by the **Anthropic Claude API**, enabling targeted accent coaching at scale
- Engineered an **AI audio pipeline**: integrated **ElevenLabs** TTS for native-speaker synthesis and STT for transcription, then routed transcriptions into Claude for accent comparison and structured improvement feedback
- Modeled a **PostgreSQL** schema via **Prisma** across 6 tables; implemented credential auth with **NextAuth.js** and bcrypt; built a user dashboard tracking XP, streaks, session history, and achievements

PawPath: AI Dog Walking App | *Python*

1st Place – RamHack 2026

- Engineered a **FastAPI** backend with **8+ RESTful endpoints** integrating **Gemini AI** for photo-based breed identification across **200+ breeds** and **Google Maps Platform** (Routes, Places, Weather) for dynamic route generation
- Built a real-time safety scoring system processing **5+ environmental signals** (temperature, AQI, UV index, humidity, wind) to assess walk conditions and surface health risk alerts per breed
- Implemented full CRUD with **Supabase** for dog profiles and walk history; integrated live GPS tracking and breed-specific exercise recommendation engine

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL (PostgreSQL), Go, Rust

AI & ML: LLM Integration, RAG Pipelines, AI Agent Design, Prompt & Context Engineering, Reinforcement Learning, RLHF, PPO, DPO, Embeddings, Vector Databases, Fine-Tuning (LoRA/QLoRA), Diffusion Models, Multimodal AI

Frameworks & Libraries: PyTorch, JAX, HuggingFace Transformers, LangChain, LangGraph, OpenAI/Anthropic SDKs

Backend & Infrastructure: Node.js, FastAPI, Express, PostgreSQL, Redis, Prisma, Docker, Nginx, JWT, OAuth 2.0, WebSockets

Tools & Platforms: Git, GitHub Actions, Railway, AWS, Vercel, Supabase, Linux, OpenTelemetry